

Project-01: Building Scalable, Secure, and High-Performance Web Applications with AWS 3-Tier Architecture

Working Application Link (Public ALB DNS)

<http://alb-web-server-134995308.ap-south-1.elb.amazonaws.com/>

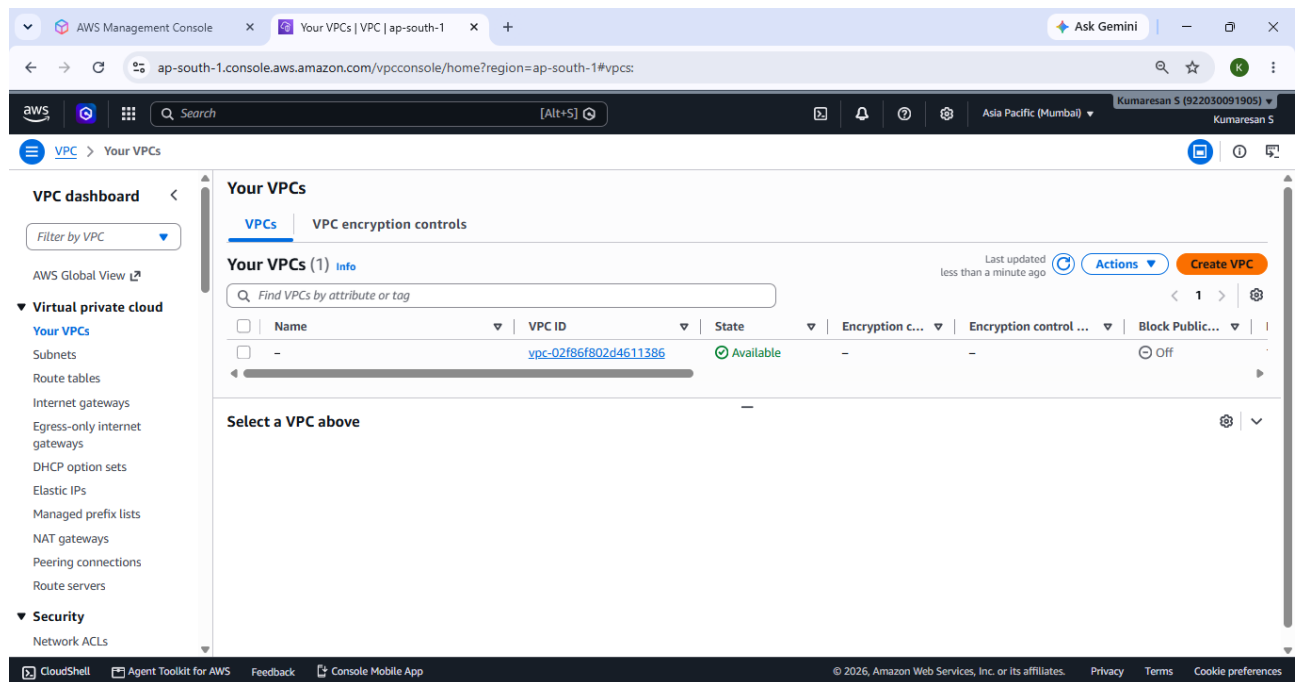
Task 1 : Design and Configure the AWS Networking Infrastructure

Create the networking components required for the application.

Tasks Create the following AWS resources:

- Custom VPC
- Internet Gateway
- NAT Gateway
- Elastic IP
- Route Tables
- Subnet Associations

VPC Create a VPC using an appropriate CIDR block.



VPC

>

Your VPCs

>

Create VPC

☐ IPAM-allocated IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy

Info

Default

VPC encryption control (\$)

Info

Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None

☐ Monitor mode
See which resources in your VPC are unencrypted but allow the creation of unencrypted resources.

☐ Enforce mode
Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name

Q Guvi 3 tier Project

Remove tag

Add tag

You can add 49 more tags

Cancel

Preview code

Create VPC

The screenshot shows the AWS Management Console interface for VPCs. At the top, there's a navigation bar with 'VPC' and 'Your VPCs' tabs. The main content area is titled 'Your VPCs' and includes a filter dropdown set to 'Filter by VPC'. Below this, there's a section for 'Virtual private cloud' with a list of VPCs. The list has columns for Name, VPC ID, State, Encryption, Encryption control, and Block Public... The 'Guvi 3 tier Project' VPC is selected, showing its VPC ID as 'vpc-07b3d399b08105f00' and its state as 'Available'. Below the list, the details for this VPC are expanded, showing tabs for Details, Resource map, CIDRs, Flow logs, Tags, Related resources, and Integrations. The 'Details' tab is active, displaying the VPC ID, state (Available), and various options like 'Block Public Access' (Off), 'DNS hostnames' (Disabled), 'DNS resolution' (Enabled), and 'Tennacy default'.

Subnets

Create 6 Subnets distributed across 2 Availability Zones.

Subnet Type ->Quantity

- Public (Web Tier) ->2
- Private (Application Tier) ->2
- Private (Database Tier) ->2

Each subnet should support at least 256 IP addresses.

Subnet Name	Ipv4 Subnet CIDR Block
Public Web 1	10.0.1.0/24
Public Web 2	10.0.2.0/24
Private App 1	10.0.3.0/24
Private App 2	10.0.4.0/24
Private DB 3	10.0.5.0/24
Private DB 4	10.0.6.0/24

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'Subnets (3)' page for the 'ap-south-1' region. It lists three existing subnets: 'subnet-0152abd6a75556f7a', 'subnet-06bcedab0ea9cadd6', and 'subnet-05763d970d22ecbab', all in an 'Available' state. The bottom screenshot shows the 'Create subnet' wizard. In the 'VPC' section, the 'vpc-07b3d399b08105f00' is selected. In the 'Subnet settings' section, 'Subnet 1 of 1' is being configured with the name 'my-subnet-01'.

Subnets (3)

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
-	subnet-0152abd6a75556f7a	Available	vpc-02f86f802d4611386	Off	172.31.32.0/2
-	subnet-06bcedab0ea9cadd6	Available	vpc-02f86f802d4611386	Off	172.31.16.0/2
-	subnet-05763d970d22ecbab	Available	vpc-02f86f802d4611386	Off	172.31.0.0/20

Create subnet

VPC
VPC ID
Create subnets in this VPC.
vpc-07b3d399b08105f00 (Guvi 3 tier Project)
vpc-02f86f802d4611386 (default)
vpc-07b3d399b08105f00 (Guvi 3 tier Project)

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
my-subnet-01
The name can be up to 256 characters long.

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

Kumaresan S

VPC

Subnets

Create subnet

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Public Web 1

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.1.0/24

256 IPs

Tags - optional

Key

Value - optional

Public Web 1

Remove

Add new tag

You can add 49 more tags.

Remove

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Search

[Alt+S]

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

Kumaresan S

VPC

Subnets

Create subnet

Availability zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.6.0/24

256 IPs

Tags - optional

Key

Value - optional

Private DB 4

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

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Asia Pacific (Mumbai)

Kumaresan S (922030091905)

Kumaresan S

VPC

Subnets

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

You have successfully created 6 subnets: subnet-098fe46a2f49231c0, subnet-0f4abe6b299eb53af, subnet-0e4f8df1c6e45c09f, subnet-0f571e5193641f12f, subnet-0c7fd57851a402f6b, subnet-07dd53ec11a512720

Last updated less than a minute ago

Actions

Create subnet

Subnets (9)

Info

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	Private App 2	subnet-0f571e5193641f12f	Available	vpc-07b3d399b08105f00 Guvi...	Off	10.0.4.0/24
☐	Private App 1	subnet-0e4f8df1c6e45c09f	Available	vpc-07b3d399b08105f00 Guvi...	Off	10.0.3.0/24
☐	Public Web 2	subnet-0f4abe6b299eb53af	Available	vpc-07b3d399b08105f00 Guvi...	Off	10.0.2.0/24
☐	Private DB 4	subnet-07dd53ec11a512720	Available	vpc-07b3d399b08105f00 Guvi...	Off	10.0.6.0/24
☐	Public Web 1	subnet-098fe46a2f49231c0	Available	vpc-07b3d399b08105f00 Guvi...	Off	10.0.1.0/24
☐	Private DB 3	subnet-0c7fd57851a402f6b	Available	vpc-07b3d399b08105f00 Guvi...	Off	10.0.5.0/24

Select a subnet

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Internet Gateway

Create an Internet Gateway.

AWS Management Console | Create internet gateway | VPC | + | Ask Gemini

ap-south-1.console.aws.amazon.com/vpconsole/home?region=ap-south-1#CreateInternetGateway:

VPC > Internet gateways > Create internet gateway

Create internet gateway info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Guvi Project IGW

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q Guvi Project IGW X Remove

Add new tag

You can add 49 more tags.

Cancel Create internet gateway

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AWS Management Console | Internet gateway details | VPC | + | Ask Gemini

ap-south-1.console.aws.amazon.com/vpconsole/home?region=ap-south-1#InternetGateway:internetGatewayId=igw-02222ee699777bd91

VPC > Internet gateways > igw-02222ee699777bd91

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs

The following internet gateway was created: igw-02222ee699777bd91 - Guvi Project IGW. You can now attach to a VPC to enable the VPC to communicate with the internet. [Attach to a VPC](#)

igw-02222ee699777bd91 / Guvi Project IGW Actions

Details info

Internet gateway ID	State	VPC ID	Owner
igw-02222ee699777bd91	Detached	-	922030091905

Tags (1)

Q Search tags

Key	Value
Name	Guvi Project IGW

Manage tags < 1 > ⚙

Attach it to the VPC.

AWS Management Console | Attach internet gateway | VPC | + | Ask Gemini

ap-south-1.console.aws.amazon.com/vpconsole/home?region=ap-south-1#AttachInternetGateway:internetGatewayId=igw-02222ee699777bd91

VPC > Internet gateways > Attach to VPC (igw-02222ee699777bd91)

The following Internet gateway was created: igw-02222ee699777bd91 - Guvi Project IGW. You can now attach to a VPC to enable the VPC to communicate with the internet. [Attach to a VPC](#)

Attach to VPC (igw-02222ee699777bd91) Info

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

Q vpc-07b3d399b08105f0d X

AWS Command Line Interface command

Cancel Attach internet gateway

Internet gateways (1) info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
Guvi Project IGW	igw-02222ee699777bd91	Attached	vpc-07b3d399b08105f00 Guvi 3 tier P...	922030091905

Select an internet gateway above

NAT Gateway

Allocate an Elastic IP.

Elastic IP addresses info

Find elastic IP addresses by attribute or tag

Name	Allocated IPv4 addr...	Type	Allocation ID	Reverse DNS record
No Elastic IP addresses found in this Region				

Select an elastic IP address

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

The screenshot shows the AWS Management Console interface for allocating an Elastic IP address. The top navigation bar includes the AWS logo, a search bar, and the user's name 'Kumaresan S'. The breadcrumb trail indicates the path: VPC > Elastic IP addresses > Allocate Elastic IP address. The main content area is titled 'Public IPv4 address pool' and lists three options: 'Amazon's pool of IPv4 addresses' (selected), 'Public IPv4 address that you bring to your AWS account with BYOIP', and 'Customer-owned pool of IPv4 addresses created from your on-premises network'. Below this, the 'Network border group' is set to 'ap-south-1'. A section for 'Global static IP addresses' is also visible. At the bottom, there is a 'Tags - optional' section. The bottom navigation bar shows 'CloudShell', 'Agent Toolkit for AWS', 'Feedback', and 'Console Mobile App'. The bottom status bar includes the copyright notice '© 2026, Amazon Web Services, Inc. or its affiliates' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
 - Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
 - Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)
 - Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group [Info](#)

Q ap-south-1

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

[Create accelerator](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tag

[Cancel](#) [Allocate](#)

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#Addresses:PublicIp=3.7.131.176

VPC > Elastic IP addresses

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs**
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs

Elastic IP address allocated successfully.
Elastic IP address 3.7.131.176 [Associate this Elastic IP address](#)

Elastic IP addresses (1/1) [Info](#)

Find elastic IP addresses by attribute or tag

Public IPv4 address: 3.7.131.176 [Clear filters](#)

☒	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
☒	Guvu EIPs	3.7.131.176	Public IP	eipalloc-0ccccfde67137db22e	-

3.7.131.176

[View IP address usage and recommendations to release unused IPs with \[Public IP insights\]\(#\)](#)

[Summary](#) [Tags](#)

Summary

Allocated IPv4 address 3.7.131.176	Type Public IP	Allocation ID eipalloc-0ccccfde67137db22e	Reverse DNS record -
Association ID -	Scope VPC	Associated instance ID -	Private IP address -

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Create a NAT Gateway in one of the Public Subnets.

Ensure private resources can access the Internet through the NAT Gateway.

ap-south-1.console.aws.amazon.com/vpconsole/home?region=ap-south-1#CreateNatGateway:

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Guvi NAT

The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.

subnet-098fe46a2f49231c0 (Public Web 1)

Connectivity type
Select a connectivity type for the NAT gateway.

☒ Public

☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

eipalloc-0cccfde67137db22e (Guvi EIPs) [Allocate Elastic IP](#)

Additional settings [Info](#)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	Guvi NAT	Remove

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create NAT gateway](#)

Route Tables

Create 3 Route Tables.

Route Table → Associated Subnets

Public Route Table → Public Subnets

Private Application Route Table → Application Subnets

Private Database Route Table → Database Subnets

Route tables (1) Info

Last updated 12 minutes ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-0687da156de6e6852	-	-	Yes	vpc-02f86f802d4611386

Select a route table

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Security

Create route table VPC | ap-sou-1

Ask Gemini

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateRouteTable:VpcId=vpc-07b3d399b08105f00

Create route table

Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

RT Public Web

VPC

The VPC to use for this route table.

vpc-07b3d399b08105f00 (Guvi 3 tier Project)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q RT Public Web X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Route tables VPC | ap-sou-1

Ask Gemini

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTables:

Route tables (5) Info

Last updated less than a minute ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-0687da156de6e6852	-	-	Yes	vpc-02f86f802d4611386
☐	RT Public Web	rtb-0df7a96e702811cfc	-	-	No	vpc-07b3d399b08105f00 G
☐	-	rtb-0e50df7a4437acab8	-	-	Yes	vpc-07b3d399b08105f00 G
☐	RT Private App	rtb-0081e7b39ecb8bf97	-	-	No	vpc-07b3d399b08105f00 G
☐	RT Private DB	rtb-01a9a14baf9f88cc0	-	-	No	vpc-07b3d399b08105f00 G

Select a route table

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Security

Route tables (1/5) Info

Last updated less than a minute ago

Actions

Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-0687da156de6e6852	-	-	Yes	vpc-02f86f802d4611386
☒ RT Public Web	rtb-0df7a96e702811cfc	-	-	No	vpc-07b3d399b08105f00
-	rtb-0e50df7a4437acab8	-	-	Yes	vpc-07b3d399b08105f00
RT Private App	rtb-0081e7b39ecb8bf97	-	-	No	vpc-07b3d399b08105f00
RT Private DB	rtb-01a9a14baf9f88cc0	-	-	No	vpc-07b3d399b08105f00

rtb-0df7a96e702811cfc / RT Public Web

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (1)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

Edit routes: rtb-0df7a96e702811cfc / RT Public Web

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

Add route

Cancel

Preview

Save changes

</

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Updated routes for rtb-0df7a96e702811cfc / RT Public Web successfully

Details

rtb-0df7a96e702811cfc / RT Public Web

Actions

Details

Route table ID

rtb-0df7a96e702811cfc

VPC

vpc-07b3d399b08105f00 | Guvi 3 tier Project

Main

No

Owner ID

922030091905

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Filter routes

Both

Edit routes

1

Destination

Target

Status

Propagated

Route Origin

0.0.0.0/0

igw-02222ee699777bd91

Active

No

Create Route

10.0.0.0/16

local

Active

No

Create Route Table

VPC dashboard

Filter by VPC

AWS Global View

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Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Route tables (1/5)

Find route tables by attribute or tag

Last updated less than a minute ago

Actions

Create route table

1

Name

Route table ID

Explicit subnet associ...

Edge associations

Main

VPC

-

rtb-0687da156de6e6852

-

-

Yes

vpc-02f86f802d4611386

☒

RT Public Web

rtb-0df7a96e702811cfc

-

-

No

vpc-07b3d399b08105f00

☐

-

rtb-0e50df7a4437acab8

-

-

Yes

vpc-07b3d399b08105f00

☐

RT Private App

rtb-0081e7b39ecb8bf97

-

-

No

vpc-07b3d399b08105f00

☐

RT Private DB

rtb-01a9a14baf9f88cc0

-

-

No

vpc-07b3d399b08105f00

rtb-0df7a96e702811cfc / RT Public Web

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Explicit subnet associations (0)

Find subnet association

Edit subnet associations

1

Name

Subnet ID

IPv4 CIDR

IPv6 CIDR

No subnet associations

You do not have any subnet associations.

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/6)

Filter subnet associations

1

Name

Subnet ID

IPv4 CIDR

IPv6 CIDR

Route table ID

☐

Private App 2

subnet-0f571e5193641f12f

10.0.4.0/24

-

Main (rtb-0e50df7a4437acab8)

☐

Private App 1

subnet-0e4f8df1c6e45c09f

10.0.3.0/24

-

Main (rtb-0e50df7a4437acab8)

☒

Public Web 2

subnet-0f4abe6b299eb53af

10.0.2.0/24

-

Main (rtb-0e50df7a4437acab8)

☐

Private DB 4

subnet-07dd53ec11a512720

10.0.6.0/24

-

Main (rtb-0e50df7a4437acab8)

☒

Public Web 1

subnet-098fe46a2f49231c0

10.0.1.0/24

-

Main (rtb-0e50df7a4437acab8)

☐

Private DB 3

subnet-0c7fd57851a402f6b

10.0.5.0/24

-

Main (rtb-0e50df7a4437acab8)

Selected subnets

subnet-098fe46a2f49231c0 / Public Web 1

subnet-0f4abe6b299eb53af / Public Web 2

Cancel

Save associations

Follow this step to create the subnet associations and routes to the RT Private App and the RT Private DB

The screenshot shows the AWS Management Console for the 'ap-south-1' region. The 'Route tables' page is displayed, showing a list of 5 route tables. The table includes columns for Name, Route table ID, Explicit subnet associations, Edge associations, Main status, and VPC. The route tables listed are:

Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC
-	rtb-0687da156de6e6852	-	-	Yes	vpc-02f86f802d4611386
RT Public Web	rtb-0df7a96e702811cfc	2 subnets	-	No	vpc-07b3d399b08105f00
-	rtb-0e50df7a4437acab8	-	-	Yes	vpc-07b3d399b08105f00
RT Private App	rtb-0081e7b39ecb8bf97	2 subnets	-	No	vpc-07b3d399b08105f00
RT Private DB	rtb-01a9a14baf9f88cc0	2 subnets	-	No	vpc-07b3d399b08105f00

Below the table, there is a section titled 'Select a route table'.

Task 2 - Deploy the Web Tier (Presentation Layer)

Deploy the Presentation Layer.

Tasks

- Launch EC2 instances in the Public Subnets.
- Configure the required Security Group.
- Connect to the instance.
- Install Nginx.
- Copy the provided HTML files.
- Verify the web page is accessible.

The screenshot shows the AWS Management Console for the 'ap-south-1' region. The 'Instances' page is displayed, showing a list of EC2 instances. The table includes columns for Name, Instance ID, Instance state, Instance type, Status check, Availability Zone, and Public IPv4 DNS. The table is currently empty, with a message stating 'No instances' and 'You do not have any instances in this region'. Below the table, there are sections for 'All instances', 'Instance alarm states', 'Service health', and 'Health events affecting EC2 instances'.

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Search [Alt+S] Ask Amazon Q

Asia Pacific (Mumbai) Kumaresan S (922030091905) Kumaresan S

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

NamePublic Web Server 1Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances1

Software Image (AMI)Canonical, Ubuntu, 26.04, amd64...read moreami-01a00762f46d584a1

Virtual server type (instance type)t3.micro

Firewall (security group)New security group

Storage (volumes)1 volume(s) - 8 GiB

CancelLaunch instancePreview code

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Search [Alt+S] Ask Amazon Q

Asia Pacific (Mumbai) Kumaresan S (922030091905) Kumaresan S

EC2 > Instances > Launch an instance

Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume TypeFree tier eligible

ami-01a00762f46d584a1 (64-bit (x86)) / ami-0b40571b9c2387b15 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 26.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Canonical, Ubuntu, 26.04, amd64 resolute image

ArchitectureAMI IDCreation DateUsernameVerified provider

64-bit (x86)ami-01a00762f46d584a12026-06-04ubuntu

Instance type

Instance type

t3.microFree tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

All generationsCompare instance types

Additional costs apply for AMIs with pre-installed software

Summary

Number of instances1

Software Image (AMI)Canonical, Ubuntu, 26.04, amd64...read moreami-01a00762f46d584a1

Virtual server type (instance type)t3.micro

Firewall (security group)New security group

Storage (volumes)1 volume(s) - 8 GiB

CancelLaunch instancePreview code

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Search [Alt+S] Ask Amazon Q

Asia Pacific (Mumbai) Kumaresan S (922030091905) Kumaresan S

EC2 > Instances > Launch an instance

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

WebAppTierCreate new key pair

Network settings

VPC - required

vpc-07b3d399b08105f00 (Guvi 3 tier Project)10.0.0.0/16

Subnet

subnet-098fe46a2f49231c0Public Web 1

VPC: vpc-07b3d399b08105f00 Owner: 922030091905

Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone

IP addresses available: 250 CIDR: 10.0.1.0/24

Create new subnet

Auto-assign public IP

Enable

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Summary

Number of instances1

Software Image (AMI)Canonical, Ubuntu, 26.04, amd64...read moreami-01a00762f46d584a1

Virtual server type (instance type)t3.micro

Firewall (security group)New security group

Storage (volumes)1 volume(s) - 8 GiB

CancelLaunch instancePreview code

EC2 > Instances > Launch an instance

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

Common security groups

sg-06218c22b2e902ec2

Compare security group rules

Advanced network configuration

Configure storage

1x 8 GiB gp3

Root volume, 3000 IOPS, Not encrypted

Add new volume

Summary

Number of instances1

Software Image (AMI)Canonical, Ubuntu, 26.04, amd64...read more

Virtual server type (instance type)t3.micro

Firewall (security group)SG Public Web

Storage (volumes)1 volume(s) - 8 GiB

CancelLaunch instancePreview code

Instances | EC2 | ap-south-1

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Public Web Se...	i-03d46124e6ab07a69	Running	t3.micro	Initializing	ap-south-1a	-

i-03d46124e6ab07a69 (Public Web Server 1)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-03d46124e6ab07a69

Public IPv4 address

3.111.198.237 | open address

Private IPv4 addresses

10.0.1.114

Instance state

Running

Public DNS

-

Instances | EC2 | ap-south-1

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Public Web Se...	i-03d46124e6ab07a69	Running	t3.micro	Initializing	ap-south-1a	-

i-03d46124e6ab07a69 (Public Web Server 1)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-03d46124e6ab07a69

Public IPv4 address

3.111.198.237 | open address

Private IPv4 addresses

10.0.1.114

Instance state

Running

Public DNS

-

Instances | EC2 | ap-south-1

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Public Web Se...	i-03d46124e6ab07a69	Running	t3.micro	Initializing	ap-south-1a	-

i-03d46124e6ab07a69 (Public Web Server 1)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-03d46124e6ab07a69

Public IPv4 address

3.111.198.237 | open address

Private IPv4 addresses

10.0.1.114

Instance state

Running

Public DNS

-

Connect to Linux instance

Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an SSH client.

Instance info

Instance

i-03d46124e6ab07a69 (Public Web Server 1)

VPC

vpc-07b3d399b08105f00

Subnet

subnet-098fe46a2f49231c0

Security groups

sg-06218c22b2e902ec2 (SG Public Web)

In web browser

In SSH client

Choose how to connect

Public subnet access

Managed SSH keys

EC2 Instance Connect

Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.

Private subnet access

Managed SSH keys

EC2 Instance Connect Endpoint

Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.

Maximum security

IAM authentication

SSM Session Manager

Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.

Direct

Troubleshoot

EC2 Serial Console

Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.

Settings

Choose public IP address

IPv4 address (3.111.198.237)

CancelConnect assistConnect

https://ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh?region=a...

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```
aws
Search [Alt+S]

* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Mon Sep 21 06:26:38 UTC 2026

System load: 0.14      Temperature: -273.1 C
Usage of /: 30.5% of 6.61GB Processes: 119
Memory usage: 26%     Users logged in: 0
Swap usage: 0%        IPv4 address for ens5: 10.0.1.114

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-1-114:~$
```

i-03d46124e6ab07a69 (Public Web Server 1)

PublicIPs: 3.111.198.237 PrivateIPs: 10.0.1.114

```
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
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aws
Search [Alt+S]

ubuntu@ip-10-0-1-114:~$ sudo apt install nginx -y
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fogiwrap nginx-doc ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 2
  Download size: 666 kB
  Space needed: 1880 kB / 4284 MB available

Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx-common all 1.28.3-2ubuntu1.11 [37.9 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx amd64 1.28.3-2ubuntu1.11 [628 kB]
Fetched 666 kB in 0s (28.8 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 124169 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.28.3-2ubuntu1.11_all.deb ...
Unpacking nginx-common (1.28.3-2ubuntu1.11) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.28.3-2ubuntu1.11_amd64v3.deb ...
Unpacking nginx (1.28.3-2ubuntu1.11) ...
```

i-03d46124e6ab07a69 (Public Web Server 1)

PublicIPs: 3.111.198.237 PrivateIPs: 10.0.1.114

```
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
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aws
Search [Alt+S]

ubuntu@ip-10-0-1-114:~$ sudo systemctl status nginx
● nginx.service A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-09-21 06:36:33 UTC; 35s ago
     Invocation: 48a74b305677472292255cc5b2665d7b
       Docs: man:nginx(8)
    Process: 27763 ExecStartPre=/usr/sbin/nginx -t -g -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 27765 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 27794 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 43ms
    CGroup: /system.slice/nginx.service
            └─27794 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─27797 "nginx: worker process"
                └─27798 "nginx: worker process"

Sep 21 06:36:33 ip-10-0-1-114 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Sep 21 06:36:33 ip-10-0-1-114 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.

ubuntu@ip-10-0-1-114:~$
```

i-03d46124e6ab07a69 (Public Web Server 1)

PublicIPs: 3.111.198.237 PrivateIPs: 10.0.1.114

The screenshot displays the AWS Management Console interface. The top section shows a terminal window with the following commands and output:

```
ubuntu@ip-10-0-1-114:~$ sudo systemctl start nginx
ubuntu@ip-10-0-1-114:~$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
ubuntu@ip-10-0-1-114:~$
```

The bottom section shows the 'Instances (1/1)' page for the EC2 instance 'i-03d46124e6ab07a69 (Public Web Server 1)'. The instance is in the 'Running' state, with a public IPv4 address of 3.111.198.237 and private IPv4 addresses of 10.0.1.114. The instance type is 't3.micro' and the status check is 'Initializing'.

The instance details page shows the following information:

- Instance ID: i-03d46124e6ab07a69
- Public IPv4 address: 3.111.198.237 | open address
- Private IPv4 addresses: 10.0.1.114
- Instance state: Running
- Public DNS: -

The bottom section shows a browser window displaying the 'Welcome to nginx!' page, which is the output of the curl command executed in the terminal.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Update the server

Run:

`sudo apt update`

`sudo apt upgrade -y`

Install Nginx

Run:

```
sudo apt install nginx -y
```

Check that Nginx is running:

```
sudo systemctl status nginx
```

Also run:

```
sudo systemctl enable nginx
```

and:

```
sudo systemctl start nginx
```

Nginx website files are stored here:

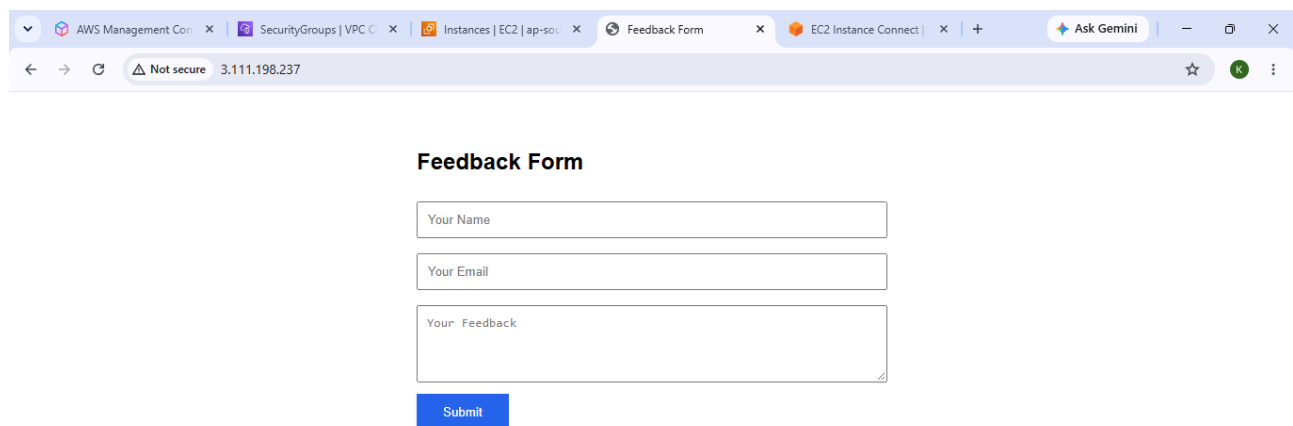
```
/var/www/html/
```

First remove the default page:

```
sudo rm /var/www/html/index.nginx-debian.html
```

Now create your HTML file:

```
sudo nano /var/www/html/index.html
```



The screenshot shows a web browser window with multiple tabs. The active tab is titled "Feedback Form" and shows a simple web form. The browser's address bar indicates the URL is "3.111.198.237" and the connection is "Not secure". The form itself is titled "Feedback Form" and contains three input fields: "Your Name", "Your Email", and "Your Feedback". Below these fields is a blue "Submit" button.

Task 3 –

Deploy the Application Tier (Backend API)

Deploy the Backend API.

Tasks

- Launch EC2 instances in the Private Application Subnets.
- Connect through the Bastion Host (if used) or Session Manager.
- Install Python 3.
- Install Flask and required Python packages.
- Check App.md in github for instructions.
- Copy app.py file from given repository.
- Install dependencies.
- Run the Flask application on Port 5000.
- Verify the application is running.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

Private App Server 1

Additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Summary

Number of instances1

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...[read more](#)
ami-01a00762f46d584a1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume Type

Free tier eligible

ami-01a00762f46d584a1 (64-bit (x86)) / ami-0b40571b9c2387b15 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 26.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 26.04, amd64 resolute image

Architecture

AMI ID

Creation Date

Username

Verified provider

64-bit (x86)

ami-01a00762f46d584a1

2026-06-04

ubuntu

Instance type

Info

Get advice

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

Additional costs apply for AMIs with pre-installed software

Summary

Number of instances1

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...[read more](#)
ami-01a00762f46d584a1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

WebAppTier

Create new key pair

Network settings

Info

VPC - required

Info

vpc-07b3d399b08105f00 (Guvi 3 tier Project)

10.0.0.0/16

Subnet

Info

subnet-0e4f8df1c6e45c09f

Private App 1

Create new subnet

Auto-assign public IP

Info

Disable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Summary

Number of instances1

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...[read more](#)
ami-01a00762f46d584a1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

EC2 > Instances > Launch an instance

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
Select security groups [Compare security group rules](#)

SG Private App sg-07950d783042a90b8 [X](#)
VPC vpc-07b3d399b08105f00
Security groups that you add or remove here will be added to or removed from all your network interfaces.

▶ **Advanced network configuration**

▼ **Configure storage** [Info](#) [Advanced](#)
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager

Summary
Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...[read more](#)
ami-01a00762f46d584a1

Virtual server type (instance type)
t3.micro

Firewall (security group)
SG Private App

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#ConnectToInstance:instanceId=i-053df0b51b1422be9

EC2 > Instances > i-053df0b51b1422be9 > Connect to instance

Connect to Linux instance [Info](#)
Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an SSH client.

Instance info

Instance i-053df0b51b1422be9 (Private App Server 1) State: Running Status: 0/2 checks passed	VPC vpc-07b3d399b08105f00	Subnet subnet-0e4f8df1c6e45c09f	Security groups sg-07950d783042a90b8 (SG Private App)
--	-------------------------------------	---	---

In web browser **In SSH client**

Choose how to connect

☐ Public subnet access ☐ Managed SSH keys
EC2 Instance Connect
Use for instances with public IPs or accessible via standard routing. Requires network connectivity to your instance. Connect using temporary SSH keys managed by AWS.

☐ Private subnet access ☐ Managed SSH keys
EC2 Instance Connect Endpoint
Use for instances in private subnets. Traffic is routed through a VPC endpoint. Connect using temporary SSH keys managed by AWS.

☒ Maximum security ☐ IAM authentication
SSM Session Manager
Uses IAM-based authentication. Connect without SSH keys, bastion hosts, or opening any inbound ports.

☐ Direct ☐ Troubleshoot
EC2 Serial Console
Access your instance's serial port for troubleshooting boot, network, and OS issues. Works without network connectivity. Requires account-level enablement.

SSM agent info
Ping status: Online

IAM role
instanceRole

[Cancel](#) [Connect assist](#) [Connect](#)

https://ap-south-1.console.aws.amazon.com/systems-manager/session-manager/i-053df0b51b1422be9?region=ap-south-1#:

Session ID: root-knsy44a7q2xsbkxd3vuj92p4 [Shortcuts](#) Instance ID: i-053df0b51b1422be9 (Private App Server 1) [Terminate session](#)

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
(env) $ nano app.py
(env) $ cat app.py
from flask import Flask, request, jsonify
import pymysql

app = Flask(__name__)

DB_CONFIG = {
    'host': 'RDS-ENDPOINT', # Replace after Step 8
    'user': 'admin',
    'password': 'YourPassword123!',
    'db': 'feedbackdb',
    'cursorclass': pymysql.cursors.DictCursor
}

@app.route('/')
def health():
    return jsonify({'status': 'Flask is running'})

@app.route('/submit', methods=['POST'])
def submit():
    data = request.get_json()
    conn = pymysql.connect(**DB_CONFIG)
    with conn.cursor() as cur:
        cur.execute(
            'INSERT INTO feedback (name, email, feedback) VALUES (%s, %s, %s)',
            (data['name'], data['email'], data['feedback'])
        )
    conn.commit()
    conn.close()
    return jsonify({'message': 'Feedback submitted successfully!'})

if __name__ == '__main__':
```

12:58 21-09-2026

The screenshot displays the AWS Systems Manager Session Manager interface. The top bar shows the session ID: root-knsy44a7q2xsbkjd3vujv92p4 and the instance ID: i-053df0b51b1422be9 (Private App Server 1). The terminal window shows the following code and output:

```
'user': 'admin',
'password': 'YourPassword123!',
'db': 'feedbackdb',
'cursorclass': pymysql.cursors.DictCursor
})

@app.route('/')
def health():
    return jsonify({'status': 'Flask is running'})

@app.route('/submit', methods=['POST'])
def submit():
    data = request.get_json()
    conn = pymysql.connect(**DB_CONFIG)
    with conn.cursor() as cur:
        cur.execute(
            'INSERT INTO feedback (name, email, feedback) VALUES (%s, %s, %s)',
            (data['name'], data['email'], data['feedback'])
        )
    conn.commit()
    conn.close()
    return jsonify({'message': 'Feedback submitted successfully!'})

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)

(venv) $ python3 app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.0.3.105:5000
Press CTRL+C to quit
```

The bottom part of the screenshot shows the terminal output after running the curl command:

```
$ curl localhost:5000
{"status": "Flask is running"}
$
```

First connect to the App Server through **AWS Systems Manager Session Manager**.
`cd /home/ssm-user`

Install pip:
`sudo apt install python3-pip -y`

Install Python virtual environment support:
`sudo apt update`
`sudo apt install python3-venv python3-full -y`

Install the required Python packages:
`pip install Flask PyMySQL gunicorn`

Install the MySQL client to connect to the Amazon RDS database:
`sudo apt install default-mysql-client -y`

Verify the installation:
`mysql --version`

Activate the virtual environment:
`. venv/bin/activate`

nano app.py

Run the Flask application:

python3 app.py

You should get output similar to:

Running on all addresses (0.0.0.0)

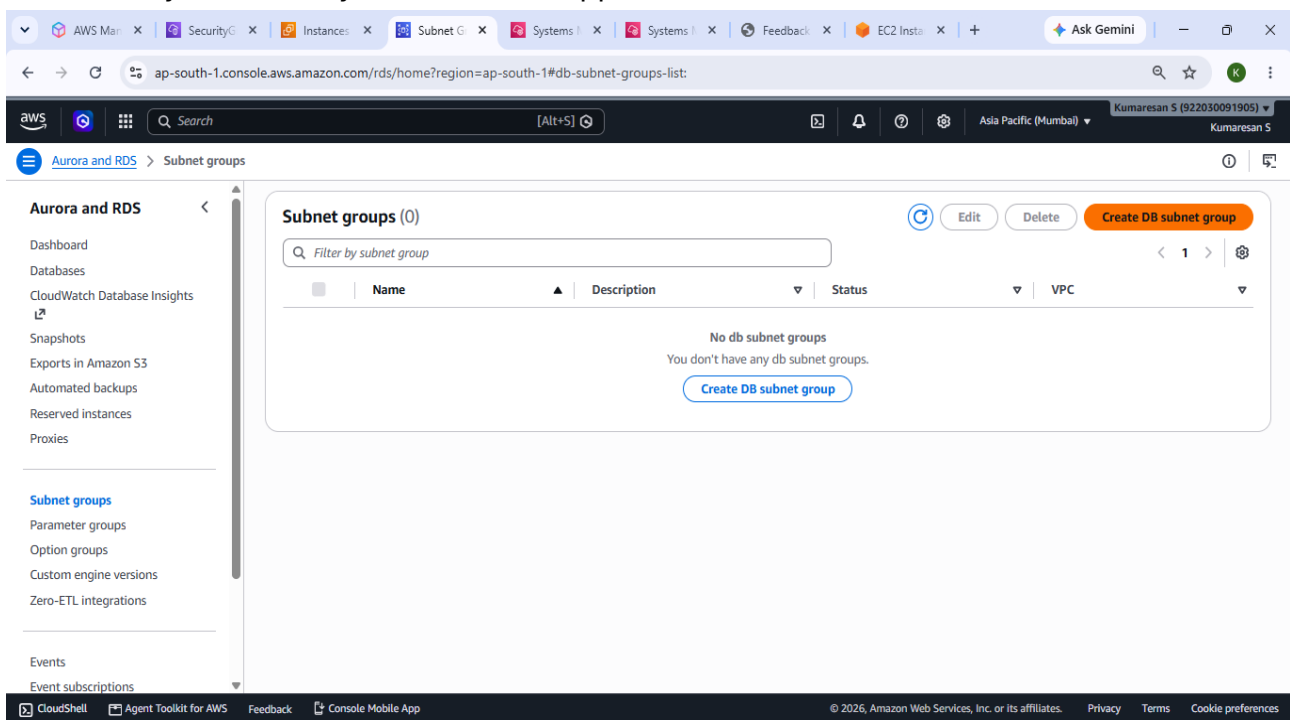
Running on <http://127.0.0.1:5000>

Task 4 - Deploy and Configure the Database Tier

Deploy the Database Layer.

Tasks

- Create an Amazon RDS MySQL instance.
- Deploy the database in the Database Subnets.
- Configure database credentials.
- Check DB.md in Github for instructions.
- Connect DB in App Ec2 terminal.
- Execute the provided SQL commands in the terminal.
- Verify connectivity from the Flask application.



ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#create-db-subnet-group:

Search

[Alt+S]

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Aurora and RDS

Subnet groups

Create DB subnet group

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.

DB-Subnet-Group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

DB-Subnet-Group

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

Guvu 3 tier Project (vpc-07b3d399b08105f00)
6 Subnets, 2 Availability Zones

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

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ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#create-db-subnet-group:

Search

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Aurora and RDS

Subnet groups

Create DB subnet group

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

ap-south-1a Xap-south-1b X

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

Private DB 3
Subnet ID: subnet-0c7fd57851a402f6b CIDR: 10.0.5.0/24

Private DB 4
Subnet ID: subnet-07dd53ec11a512720 CIDR: 10.0.6.0/24

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet name	Subnet ID	CIDR block
ap-south-1a	Private DB 3	subnet-0c7fd57851a402f6b	10.0.5.0/24
ap-south-1b	Private DB 4	subnet-07dd53ec11a512720	10.0.6.0/24

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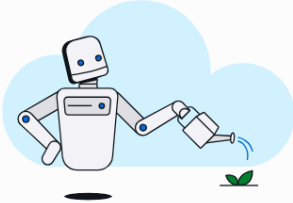
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ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#databases:

Successfully created DB-Subnet-Group. [View subnet group](#)

Databases (0)

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region	Size
 No resources No resources for efficient						

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ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible) 

☐ Aurora (PostgreSQL Compatible) 

☒ MySQL 

☐ PostgreSQL 

☐ MariaDB 

☐ Oracle 

☐ Microsoft SQL Server 

☐ IBM Db2 

Choose a database creation method

☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

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ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

DB instance identifier

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

three-tier-mysql

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Master username

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password

[Info](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password

[Info](#)

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Availability and durability

Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

Multi-AZ DB cluster deployment (3 instances)	Multi-AZ DB instance deployment (2 instances)	Single-AZ DB instance deployment (1 instance)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:	Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:	Creates a single DB instance without standby instances. This setup provides:
+ 99.95% uptime + Redundancy across Availability Zones + Increased read capacity + Reduced write latency	+ 99.95% uptime + Redundancy across Availability Zones	+ 99.5% uptime + No data redundancy
The diagram shows a dashed box labeled "Write/read endpoint" containing a cylinder icon labeled "AZ 1" and "Primary instance + SSD". A double-headed arrow connects it to another dashed box containing two cylinder icons labeled "AZ 2" and "AZ 3", both labeled "Readable standby + SSD".	The diagram shows a dashed box labeled "Write/read endpoint" containing a cylinder icon labeled "AZ 1" and "Primary instance". A double-headed arrow connects it to another dashed box labeled "Standby (no endpoint)" containing a cylinder icon labeled "AZ 2" and "Standby".	The diagram shows a dashed box labeled "Write/read endpoint" containing a cylinder icon labeled "AZ 1" and "Primary instance".

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AWS MarketplaceSecurity GroupsInstancesCreate databaseSystemsSystemsFeedbackEC2 Instance Profiles

Ask Gemini

ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#launch-dbinstance:

Search[Alt+S]

Asia Pacific (Mumbai)Kumaresan S (922030091905)Kumaresan S

Aurora and RDSDatabasesCreate database

1Help

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength Very strong
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

▼ **Additional credentials settings**

Database authentication options [Info](#)

Connect to the Amazon RDS Instance

From the private EC2 instance, connect to the MySQL database:

```
mysql -h <RDS-ENDPOINT> -u admin -p
```

Enter the RDS master password when prompted.

Create the Database

Inside the MySQL shell, create the application database:

```
CREATE DATABASE feedbackdb;
```

Select the database:

```
USE feedbackdb;
```

Create the Feedback Table

Create the table to store customer feedback:

```
CREATE TABLE feedback (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  email VARCHAR(100),  
  feedback TEXT,  
  created TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Verify the Database

Check that the table has been created successfully:

```
SHOW TABLES;
```

View the table structure:

```
DESCRIBE feedback;
```

Configure the Flask Application

Open the Flask application:

```
nano ~/app.py
```

Locate the database configuration:

```
'host': 'RDS-ENDPOINT',
```

Replace it with your actual Amazon RDS details:

```
db = pymysql.connect(  
  host='YOUR-RDS-ENDPOINT',  
  user='admin',  
  password='YourPassword123!',  
  database='feedbackdb'  
)
```

Replace **YOUR-RDS-ENDPOINT** with the endpoint of your Amazon RDS instance.

Restart the Flask Application

Stop the currently running Flask application:

```
pkill -f "python3 app.py"
```

Start the application again:

```
python3 ~/app.py &
```

Test Database Connectivity

Send a test request to the Flask API:

```
curl -X POST http://localhost:5000/submit \
-H "Content-Type: application/json" \
-d '{
  "name": "Test",
  "email": "t@t.com",
  "feedback": "Hello RDS!"
}'
```

Expected Response

```
{
  "message": "Feedback submitted successfully!"
}
```

Verify the Data in Amazon RDS

Reconnect to the database:

```
mysql -h <RDS-ENDPOINT> -u admin -p
```

Select the database:

```
USE feedbackdb;
```

View the stored feedback:

```
SELECT * FROM feedback;
```

If the record appears, the Flask application is successfully connected to the Amazon RDS database.

Task 5 - Implement Network and Instance Security

Create Security Groups with appropriate inbound and outbound rules.

Configure Security Groups for:

- Web Tier
- Application Tier
- Database Tier
- Load Balancer

Ensure only the required ports are allowed between tiers.

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SecurityGroups:

aws

VPC > Security Groups

gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

▼ Security

Network ACLs

Security groups

▼ PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

Service networks

Lattice services

Resource configurations

Resource gateways

Target groups

Security Groups (2) info

Find security groups by attribute or tag

	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-0b4e9e4002c160aa1	default	vpc-07b3d399b08105f00	default VPC secu
☐	-	sg-0147200b92d1bb1bc3	default	vpc-02f86f802d4611386	default VPC secu

Select a security group

Create security group

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSecurityGroup:VpcId=vpc-07b3d399b08105f00

aws

VPC > Security Groups > Create security group

Create security group info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name info

SG Public Web

Name cannot be edited after creation.

Description info

SG Public Web

VPC info

vpc-07b3d399b08105f00 (Guvi 3 tier Project)

Inbound rules info

Type	Protocol	Port range	Source	Description - optional	
HTTP	TCP	80	Anyw...		Delete
			0.0.0.0/0		X
SSH	TCP	22	My IP		Delete
			223.239.59.206/32		X

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSecurityGroup:VpcId=vpc-07b3d399b08105f00

aws

VPC > Security Groups > Create security group

SSH

TCP

22

Custom

pl-0fa83cebf909345ca

pl-0fa83cebf909345ca

Delete

Add rule

Outbound rules info

Type	Protocol	Port range	Destination	Description - optional	
All traffic	All	All	Custom		Delete
			0.0.0.0/0		X

Add rule

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel

Create security group

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSecurityGroup:VpcId=vpc-07b3d399b08105f00

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

Kumaresan S

VPC > Security Groups > Create security group

Create security group

Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name | Info

SG ALB

Name cannot be edited after creation.

Description | Info

SG ALB

VPC | Info

vpc-07b3d399b08105f00 (Guvi 3 tier Project)

Inbound rules

Info

Type | Info

Protocol | Info

Port range | Info

Source | Info

Description - optional | Info

HTTP

TCP

80

Anyw...

0.0.0.0/0

sg-06218c22b2e902ec

sg-06218c22b2e902ec2

Custom TCP

TCP

5000

Custom

sg-06218c22b2e902ec

sg-06218c22b2e902ec2

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSecurityGroup:VpcId=vpc-07b3d399b08105f00

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

Kumaresan S

VPC > Security Groups > Create security group

Custom TCP

TCP

5000

Custom

sg-06218c22b2e902ec

sg-06218c22b2e902ec2

Add rule

Outbound rules

Info

Type | Info

Protocol | Info

Port range | Info

Destination | Info

Description - optional | Info

All traffic

All

All

Custom

0.0.0.0/0

Add rule

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel Create security group

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSecurityGroup:VpcId=vpc-07b3d399b08105f00

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

Kumaresan S

VPC > Security Groups > Create security group

Create security group

Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name | Info

SG Private App

Name cannot be edited after creation.

Description | Info

SG Private App

VPC | Info

vpc-07b3d399b08105f00 (Guvi 3 tier Project)

Inbound rules

Info

Type | Info

Protocol | Info

Port range | Info

Source | Info

Description - optional | Info

Custom TCP

TCP

5000

Custom

sg-0eb078461b9ce06d4

sg-0eb078461b9ce06d4

Add rule

Use: "sg-0eb078461b9ce06d4"

CIDR blocks

Security Groups

SG ALB | sg-0eb078461b9ce06d4

SG ALB

Prefix lists

sg-0eb078461b9ce06d4

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSecurityGroup:VpcId=vpc-07b3d399b08105f00

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

VPC > Security Groups > Create security group

sg-0eb078461b9ce06d4

Add rule

Outbound rules

Type

All traffic

Protocol

All

Port range

All

Destination

Custom

0.0.0.0/0

Delete

Add rule

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel Create security group

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#CreateSecurityGroup:VpcId=vpc-07b3d399b08105f00

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

VPC > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

SG Private DB

Description

SG Private DB

VPC

vpc-07b3d399b08105f00 (Guvi 3 tier Project)

Inbound rules

Type

MYSQL/Aurora

Protocol

TCP

Port range

3306

Source

Custom

sg-07950d783042a90b8

sg-07950d783042a90b8

Add rule

The image shows two screenshots of the AWS Management Console. The top screenshot is the 'Create security group' wizard. It shows the 'Outbound rules' section with a rule for 'All traffic' to '0.0.0.0/0'. The bottom screenshot shows the 'Security Groups' list with 6 groups.

Name	Security group ID	Security group name	VPC ID	Description
-	sg-0b4e9e4002c160aa1	default	vpc-07b3d399b08105f00	default VPC security group
SG Private DB	sg-05c4f28ce58bd5b9f	SG Private DB	vpc-07b3d399b08105f00	SG Private DB
SG ALB	sg-0eb078461b9ce06d4	SG ALB	vpc-07b3d399b08105f00	SG ALB
-	sg-0147200b92d1b1bc3	default	vpc-02f86f802d4611386	default VPC security group
SG Public Web	sg-06218c22b2e902ec2	SG Public Web	vpc-07b3d399b08105f00	SG Public Web
SG Private App	sg-07950d783042a90b8	SG Private App	vpc-07b3d399b08105f00	SG Private App

Task 6 - Configure the Internet-Facing Application Load Balancer

Create an internet-facing Application Load Balancer.

Tasks

- Create a Target Group.
- Register Web EC2 instances.
- Configure Health Checks.
- Attach the Load Balancer to the Public Subnets.
- Verify the application is accessible using the ALB DNS name.

Target groups

Filter target groups

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
No target groups						
You don't have any target groups in ap-south-1						

Create target group

0 target groups selected

Select a target group above.

Create target group

Protocol

HTTP

Port

80

IP address type

IPv4

IPv6

VPC

vpc-07b3d399b08105f00 (Guvi 3 tier Project)

Protocol version

HTTP1

HTTP2

gRPC

Create target group

Health checks

Health check protocol

HTTP

Health check path

/

Advanced health check settings

Health check port

Traffic port

Override

Healthy threshold

3

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Step 1

Create target group

Step 2 - *recommended*

Register targets

Step 3

Review and create

Register targets - *recommended*

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/4)

☐	Instance ID	Name	State	Security groups	Zone
☐	i-0d240ac0874d802ac	Private App Server 2	Running	SG Private App	ap-south-1b
☐	i-053df0b51b1422be9	Private App Server 1	Running	ec2-rds-1, SG Private App	ap-south-1a
☒	i-095c91397778be272	Public Web Server 2	Running	SG Public Web	ap-south-1b
☒	i-03d46124e6ab07a69	Public Web Server 1	Running	SG Public Web	ap-south-1a

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

1-65535 (separate multiple ports with commas)

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancers:

aws [Search] [Alt+S]

EC2 > Load balancers

Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

Load Balancing

- Load Balancers**
- Target Groups
- Trust Stores

Auto Scaling

- Auto Scaling Groups

Settings

Load balancers What's new?

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 > ⚙️

	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
--	------	-------	------	--------	-----------------	--------	--------------------

0 load balancers selected

Select a load balancer above.

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#SelectCreateELBWizard:

EC2 > Load balancers > Compare and select load balancer type

Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

EC2 > Load balancers > Create Application Load Balancer

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-07b3d399b08105f00 (Guvi 3 tier Project)
10.0.0.0/16

Create VPC

IP pools [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and Internet gateway (IGW) routes configured to allow traffic.

subnet-098fe46a2f49231c0 (Public Web 1)
10.0.1.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and Internet gateway (IGW) routes configured to allow traffic.

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

EC2 > Load balancers > Create Application Load Balancer

subnet-0f4abe6b299eb53af (Public Web 2)
10.0.2.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

SG ALB (sg-0eb078461b9ce06d4)
Inbound rule (2) Outbound rule (1)

Create security group

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 Remove

Protocol HTTP Port 80
1-65535

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

EC2 > Load balancers > Create Application Load Balancer

Create Application Load Balancer

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

▶ How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB-Web-Server

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Info

Scheme can't be changed after the load balancer is created.

Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type

Info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

IPv4

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroups:

EC2 > Target groups

Target groups (1/1)

Info | What's new?

Filter target groups

	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☒	TG-Web-Server	arn:aws:elasticloadbalancing...	80	HTTP	Instance	ALB-Web-Server	vpc-07b

Target group: TG-Web-Server

Target groups have requests to distributed registered targets using the protocol and port defined separately. Health checks are performed on the registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Over
☐	i-03d46124e6ab07a69	Public Web Ser...	80	ap-south-1a (a...	Healthy	-	-	-
☐	i-095c91397778be272	Public Web Ser...	80	ap-south-1b (a...	Healthy	-	-	-

The first screenshot shows the AWS Management Console for the 'Load balancers' page in the 'ap-south-1' region. The 'Load balancers (1/1)' list shows a single load balancer named 'ALB-Web-Server' in an 'Active' state, with an 'application' type and 'Internet-facing' scheme. The 'Details' section for 'ALB-Web-Server' shows it is an 'Application' type, 'Internet-facing' scheme, 'Active' status, hosted in the 'ZP97RAFLXTNZK' zone, using VPC 'vpc-07b3d399b08105f00' and subnets 'subnet-0f4abe6b299eb53af' and 'subnet-0a1a1a1a1a1a1a1a1'. It is associated with '2 Availability Zones' and was created on 'September 21, 2026, 14:14 (UTC+05:30)'.

The second screenshot shows the 'Target groups' page. The 'Target groups (1/1)' list shows a single target group named 'TG-Web-Server' with an ARN of 'arn:aws:elasticloadbalancing:ap-south-1:123456789012:targetgroup/tg-web-server/123456789012', port '80', and protocol 'HTTP'. It is an 'Instance' target type associated with the 'ALB-Web-Server' load balancer and VPC 'vpc-07b3d399b08105f00'. The 'Details' section for 'TG-Web-Server' shows 'Registered targets (2)' with a table of targets. The table has columns for 'Instance ID', 'Name', 'Port', 'Zone', 'Health status', 'Health status details', 'Admin...', and 'Over...'. The 'Anomaly mitigation' status is 'Not applicable'.

Task 7 - Configure Internal Application Load Balancer

Create an Internal Load Balancer for the Application Tier.

Tasks

- Create an Internal ALB.
- Create a Target Group.
- Register Flask EC2 instances.
- Configure Health Checks.
- Verify communication between the Web Tier and Application Tier.

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.

TG-App-Server

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.

TG-App-Server

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.

TG-App-Server

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

Protocol
Protocol for communication between the load balancer and targets.

HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

5000

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-07b3d399b08105f00 (Guvi 3 tier Project)
10.0.0.0/16

Protocol version

☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ **gRPC**

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.

TG-App-Server

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

Protocol
Protocol for communication between the load balancer and targets.

HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

5000

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-07b3d399b08105f00 (Guvi 3 tier Project)
10.0.0.0/16

Protocol version

☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ **gRPC**

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.

TG-App-Server

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

Protocol
Protocol for communication between the load balancer and targets.

HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

5000

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-07b3d399b08105f00 (Guvi 3 tier Project)
10.0.0.0/16

Protocol version

☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ **gRPC**

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/health

Up to 1024 characters allowed.

Advanced health check settings

Health check port
The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.

☒ **Traffic port**

☐ **Override**

Healthy threshold
The number of consecutive health checks successes required before considering an unhealthy target healthy.

3

2-10

Step 1 Create target group

EC2 Instance Connect

Databases | Aurora and

Ask Gemini

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:

Search

EC2 > Target groups > Create target group

Unhealthy threshold

The number of consecutive health check failures required before considering a target unhealthy.

3

2-10

Timeout

The amount of time, in seconds, during which no response means a failed health check.

5

seconds

2-120

Interval

The approximate amount of time between health checks of an individual target

30

seconds

5-300

Success codes

The HTTP codes to use when checking for a successful response from a target. You can specify multiple values (for example, "200,202") or a range of values (for example, "200-299").

200

Target optimizer - optional

Enforce a maximum number of concurrent requests on a target. This is particularly useful for compute-intensive workloads like complex data processing or inference.

Target optimizer

Step 1 Create target group

EC2 Instance Connect

Databases | Aurora and

Ask Gemini

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:

Search

EC2 > Target groups > Create target group

Target optimizer - optional

Enforce a maximum number of concurrent requests on a target. This is particularly useful for compute-intensive workloads like complex data processing or inference.

Target optimizer

Off - Default

Use standard load balancing.

On

Enforce a maximum number of concurrent requests on a target. Requires targets to have target optimizer agent installed.

Attributes

Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

Tags - optional

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel

Next

Step 2 Create target group

EC2 Instance Connect

Databases | Aurora and

Ask Gemini

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:

Search

EC2 > Target groups > Create target group

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (4)

Filter instances

	Instance ID	Name	State	Security groups	Zone
☐	i-0d240ac0874d802ac	Private App Server 2	Running	SG Private App	ap-south-1b
☐	i-053df0b51b1422be9	Private App Server 1	Running	ec2-rds-1, SG Private App	ap-south-1a
☐	i-095c91397778be272	Public Web Server2	Running	SG Public Web	ap-south-1b
☐	i-03d46124e6ab07a69	Public Web Server 1	Running	SG Public Web	ap-south-1a

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

5000

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:

EC2 > Target groups > Create target group

5000

1-65535 (separate multiple ports with commas)

Include as pending below

2 selections are now pending below. Include more or register targets when ready.

Review targets

Targets (2)

Filter targets

Show only pending

Remove all pending

< 1 >

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet
i-0d240ac0874d802ac	Private App Server 2	5000	Running	SG Private App	ap-south-1b	10.0.4.141	subnet-0...
i-053df0b51b1422be9	Private App Server 1	5000	Running	ec2-rds-1, SG Private App	ap-south-1a	10.0.3.105	subnet-0...

2 pending

Cancel Previous Next

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:

EC2 > Target groups > Create target group

Health check protocol
HTTP

Timeout
5 seconds

Health check path
/health

Healthy threshold
3

Health check port
traffic-port

Unhealthy threshold
3

Interval
30 seconds

Success codes
200

Step 2: Register targets

Edit

Targets (2)

Instance ID	Name	Port	Zone
i-053df0b51b1422be9	Private App Server 1	5000	ap-south-1a
i-0d240ac0874d802ac	Private App Server 2	5000	ap-south-1b

Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

Cancel Previous Create target group

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancers:

EC2 > Load balancers

Load balancers (1/1) What's new?

Actions

Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 >

Name	State	Type	Scheme	IP address type	VPC ID	Availability Zone
ALB-Web-Server	Active	application	Internet-facing	IPv4	vpc-07b3d399b08105f00	2 Availability Zones

Loading... (1 processed)

Load balancer: ALB-Web-Server

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

Search

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

EC2 > Load balancers > Create Application Load Balancer

Create Application Load Balancer

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

How Application Load Balancers work

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB-App-Server

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | Info
Scheme can't be changed after the load balancer is created.

☐ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☒ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type | Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

CloudShellAgent Toolkit for AWSFeedbackConsole Mobile App

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

Search

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

EC2 > Load balancers > Create Application Load Balancer

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-07b3d399b08105f00 (Guvi 3 tier Project)
10.0.0.0/16

Create VPC

IP pools | Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses
Compatible with Internet-facing scheme, IPv4 and Dualstack IP address types.

Availability Zones and subnets | Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and Internet gateway (IGW) routes configured to allow traffic.

subnet-0e4f8df1c6e45c09f (Private App 1)
10.0.3.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1)

☒ ap-south-1b (aps1-az3)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and Internet gateway (IGW) routes configured to allow traffic.

CloudShellAgent Toolkit for AWSFeedbackConsole Mobile App

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

Search

Asia Pacific (Mumbai)

Kumaresan S (922030091905)

EC2 > Load balancers > Create Application Load Balancer

☒ ap-south-1b (aps1-az3)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and Internet gateway (IGW) routes configured to allow traffic.

subnet-0f571e5193641f12f (Private App 2)
10.0.4.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1)

Security groups

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

Create security group

SG ALB (sg-0eb078461b9ce06d4)
Inbound rule (2) Outbound rule (1)

Listeners and routing

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 Remove

The screenshot displays the AWS Management Console interface for creating an Application Load Balancer. The 'Routing action' is configured to 'Forward to target groups'. The 'Target group' is 'TG-App-Server' with a weight of 1 and 100% percent. The 'Listener tags' section is optional. The 'Service integrations' section shows AWS WAF and AWS Global Accelerator. The 'Attributes' section has a note about default attributes. The 'Creation workflow and status' section shows 'Server-side tasks and status'. The 'Create load balancer' button is highlighted.

Task 8 - Configure Nginx Reverse Proxy for API Routing Update the Nginx configuration.

Tasks

- Replace the default Nginx configuration with the provided nginx.conf.
- Copy nginx.conf file from given github and replace.
- Check proxy.md in github.
- Configure Nginx to forward API requests to the Internal Application Load Balancer.
- Restart the Nginx service.
- Verify successful communication between the Web Tier and Application Tier.

1. Edit the Nginx configuration:

```
sudo nano /etc/nginx/nginx.conf
```

2. Validate the configuration:

```
sudo nginx -t
```

3. Restart Nginx:

sudo systemctl restart nginx

4. Verify Nginx:

sudo systemctl status nginx

Task 9 - Integrate the Flask Application with Amazon RDS Update the application configuration.

Tasks

- Configure database connection details.
- Verify the application connects successfully to Amazon RDS.
- Submit sample feedback through the application.
- Confirm the data is stored in the database.

Session ID: root-8jcs4oe929fvicgc9p4oqpy2hi [Shortcuts](#) Instance ID: i-053df0b51b1422be9 (Private App Server 1) [Terminate session](#)

```
$ mysql -h three-tier-mysql.cpukm8ci0ey5.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 74
Server version: 8.4.9 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE feedbackdb;
Query OK, 1 row affected (0.06 sec)

mysql> USE feedbackdb;

Database changed
mysql> CREATE TABLE feedback (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> name VARCHAR(100),
  -> email VARCHAR(100),
  -> feedback TEXT,
  -> created TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_feedbackdb |
+-----+

```

AWS Manager

VPC | ap-south-1

Instances | EC2

Systems Manager

EC2 Instance Con

Aurora and RDS

Ask Gemini

ap-south-1.console.aws.amazon.com/systems-manager/session-manager/i-053df0b51b1422be9?region=ap-south-1#;

Session ID: root-8jcs4oe929fvicgc9p4oqpy2hi [Shortcuts](#) Instance ID: i-053df0b51b1422be9 (Private App Server 1) [Terminate session](#)

```
mysql> USE feedbackdb;

Database changed
mysql> CREATE TABLE feedback (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> name VARCHAR(100),
  -> email VARCHAR(100),
  -> feedback TEXT,
  -> created TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_feedbackdb |
+-----+
| feedback |
+-----+
1 row in set (0.03 sec)

mysql> DESCRIBE feedback;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| id    | int  | NO   | PRI | NULL    | auto_increment |
| name  | varchar(100) | YES |     | NULL    |               |
| email | varchar(100) | YES |     | NULL    |               |
| feedback | text | YES |     | NULL    |               |
| created | timestamp | YES |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----+-----+-----+-----+-----+-----+
5 rows in set (0.05 sec)

mysql>
```

Session ID: root-8jcs4oe929fvcgc9p4oqpy2hi Instance ID: i-053df0b51b1422be9 (Private App Server 1)

```
python3 app.py$ (venv) $ python3 ~/app.py &
(venv) $ python3: can't open file '/home/ssm-user/app.py': [Errno 2] No such file or directory
^C
[1] + Done(2) python3 app.py
(venv) $ python3 app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://10.0.3.105:5000
Press CTRL+C to quit
10.0.3.95 - - [21/Sep/2026 09:54:59] "GET / HTTP/1.1" 200 -
10.0.4.175 - - [21/Sep/2026 09:55:08] "GET / HTTP/1.1" 200 -
10.0.3.95 - - [21/Sep/2026 09:55:29] "GET / HTTP/1.1" 200 -
10.0.4.175 - - [21/Sep/2026 09:55:38] "GET / HTTP/1.1" 200 -
10.0.3.95 - - [21/Sep/2026 09:55:59] "GET / HTTP/1.1" 200 -
10.0.4.175 - - [21/Sep/2026 09:56:08] "GET / HTTP/1.1" 200 -
^C(venv) $ python3 app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://10.0.3.105:5000
Press CTRL+C to quit
10.0.4.175 - - [21/Sep/2026 09:58:38] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [21/Sep/2026 09:58:45] "POST /submit HTTP/1.1" 200 -
10.0.3.95 - - [21/Sep/2026 09:59:00] "GET / HTTP/1.1" 200 -
10.0.4.175 - - [21/Sep/2026 09:59:08] "GET / HTTP/1.1" 200 -
10.0.3.95 - - [21/Sep/2026 09:59:30] "GET / HTTP/1.1" 200 -
10.0.4.175 - - [21/Sep/2026 09:59:38] "GET / HTTP/1.1" 200 -
10.0.3.95 - - [21/Sep/2026 10:00:00] "GET / HTTP/1.1" 200 -
```

Session ID: root-freo5u5n9y87d3xun6dkf59ptu Instance ID: i-053df0b51b1422be9 (Private App Server 1)

```
$ curl -X POST http://localhost:5000/submit \
-H "Content-Type: application/json" \
-d '{
  "name": "Test",
  "email": "t@t.com",
  "feedback": "Hello RDS!"
}' >>> >>>
{"message": "Feedback submitted successfully!"}
$
```

aws

```
ubuntu@ip-10-0-1-114:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-09-21 06:36:33 UTC; 3h 53min ago
   Invocation: 48a74b305677472292255cc5b2665d7b
     Docs: man:nginx(8)
    Main PID: 27794 (nginx)
      Tasks: 3 (limit: 627)
    Memory: 3.4M (peak: 7.3M)
       CPU: 163ms
    CGroup: /system.slice/nginx.service
            └─27794 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─27797 "nginx: worker process"
                └─27798 "nginx: worker process"

Sep 21 06:36:33 ip-10-0-1-114 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Sep 21 06:36:33 ip-10-0-1-114 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-10-0-1-114:~$ sudo cp /etc/nginx/nginx.conf /etc/nginx/nginx.conf.backup
ubuntu@ip-10-0-1-114:~$ sudo nano /etc/nginx/nginx.conf
ubuntu@ip-10-0-1-114:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-10-0-1-114:~$ sudo systemctl restart nginx
ubuntu@ip-10-0-1-114:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-09-21 10:32:24 UTC; 19s ago
   Invocation: fcd93effe01444f84892bbc684c1c7f
     Docs: man:nginx(8)
    Process: 30224 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
```

i-03d46124e6ab07a69 (Public Web Server 1)

PublicIPs: 3.111.198.237 PrivateIPs: 10.0.1.114

The screenshot displays two windows from an AWS Systems Manager session. The top window is a terminal showing the execution of a curl command to submit feedback data to a local endpoint, followed by a MySQL query to retrieve the stored data. The output shows a successful submission and a table with one row of feedback data.

```
$ curl -X POST http://localhost:5000/submit \
-H "Content-Type: application/json" \
-d '{
  "name": "Test",
  "email": "t@t.com",
  "feedback": "Hello RDS!"
}' >>>>
{"message": "Feedback submitted successfully!"}
$ mysql -h three-tier-mysql1.cpukm8ci0ey5.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 85
Server version: 8.4.9 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE feedbackdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM feedback;
+----+-----+-----+-----+-----+
| id | name | email | feedback | created |
+----+-----+-----+-----+-----+
| 1  | Test | t@t.com | Hello RDS! | 2026-09-21 09:58:45 |
+----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

The bottom window shows a web browser displaying the feedback form. The form contains input fields for name, email, and a text area for feedback, along with a submit button. Below the form, the terminal output shows the execution of commands to configure and restart Nginx, and a curl command to test the application endpoint.

```
</head>
<body>
  <h2>Feedback Form</h2>
  <input id='name' placeholder='Your Name' />
  <input id='email' placeholder='Your Email' type='email' />
  <textarea id='feedback' rows='4' placeholder='Your Feedback'></textarea>
  <button onclick='submit()'>Submit</button>
  <p id='msg'></p>
  <script>
    function submit() {
      const name = document.getElementById('name').value;
      const email = document.getElementById('email').value;
      const feedback = document.getElementById('feedback').value;
      fetch('/api/submit', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ name, email, feedback })
      })
      .then(r => r.json())
      .then(d => { document.getElementById('msg').innerText = d.message; })
      .catch(() => { document.getElementById('msg').innerText = 'Error submitting.'; });
    }
  </script>
</body>
</html>

ubuntu@ip-10-0-1-114:~$ sudo nano /etc/nginx/nginx.conf
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-10-0-1-114:~$ sudo systemctl restart nginx
ubuntu@ip-10-0-1-114:~$ curl http://internal-ALB-App-Server-1492966281.ap-south-1.elb.amazonaws.com:5000/
{"status": "Flask is running"}
```

Task 10 - Validate the Complete 3-Tier Architecture

Verify the complete application.

The learner should demonstrate:

- Web application is accessible through the Public Application Load Balancer.
- Feedback form accepts user input.
- Nginx successfully forwards API requests to the Internal Load Balancer.
- Flask application processes requests successfully.
- Feedback data is stored in Amazon RDS MySQL.
- Data can be verified using SQL queries. Security groups allow only the required traffic between application tiers.

Feedback Form

Feedback submitted successfully!

Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed

mysql> SELECT * FROM feedback;

id	name	email	feedback	created
1	Test	t@t.com	Hello RDS!	2026-09-21 09:58:45
2	Test	t@t.com	Hello RDS!	2026-09-21 10:05:47
3	Abcd	abcd123.com	The session was good	2026-09-21 10:06:07
4	Kumaresan	test@example.com	Testing through Nginx	2026-09-21 12:49:57
5	Xyz	xyz@123.com	This session is very good	2026-09-21 12:50:21
6	Guvi	guviDevops@123.com	All should done the project	2026-09-21 13:03:01
7	Arun	arun@example.com	The feedback form is working correctly.	2026-09-21 16:11:01

7 rows in set (0.00 sec)

mysql> SELECT * FROM feedback;

id	name	email	feedback	created
1	Test	t@t.com	Hello RDS!	2026-09-21 09:58:45
2	Test	t@t.com	Hello RDS!	2026-09-21 10:05:47
3	Abcd	abcd123.com	The session was good	2026-09-21 10:06:07
4	Kumaresan	test@example.com	Testing through Nginx	2026-09-21 12:49:57
5	Xyz	xyz@123.com	This session is very good	2026-09-21 12:50:21
6	Guvi	guviDevops@123.com	All should done the project	2026-09-21 13:03:01
7	Arun	arun@example.com	The feedback form is working correctly.	2026-09-21 16:11:01
8	Priya	priya@example.com	RDS database connectivity test completed successfully	2026-09-21 16:13:35

8 rows in set (0.00 sec)

Additional Task:

Configure Auto Scaling

Enhance the architecture by implementing Amazon EC2 Auto Scaling for improved scalability and fault tolerance.

=> ASG for Web server

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTemplate:

EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required

Web-Server-Launch-Template

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '\', '\'', '@'.

Template version description

Web-Server-Launch-Template

Max 255 chars

Auto Scaling guidance

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags

Source template

Summary

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...read more
ami-01a00762f46d584a1

Virtual server type (instance type)
t3.micro

Firewall (security group)
SG Public Web

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTemplate:

EC2 > Launch templates > Create launch template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

Application and OS Images (Amazon Machine Image) - required

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Recently launched

Currently in use

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

ubuntu/images/hvm-ssd-gp3/ubuntu-resolute-26.04-amd64-server-20260604
ami-01a00762f46d584a1
2026-06-04T06:47:07.000Z Architecture: 64-bit (x86) Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred

Description

Summary

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...read more
ami-01a00762f46d584a1

Virtual server type (instance type)
t3.micro

Firewall (security group)
SG Public Web

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

Application and OS Images (Amazon Machine Image) - required

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Recently launched

Currently in use

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

ubuntu/images/hvm-ssd-gp3/ubuntu-resolute-26.04-amd64-server-20260604
ami-01a00762f46d584a1
2026-06-04T06:47:07.000Z Architecture: 64-bit (x86) Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred

Description

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTemplate:

EC2 > Launch templates > Create launch template

Description

Canonical, Ubuntu, 26.04, amd64 resolute image

Architecture x86_64 AMI ID ami-01a00762f46d584a1 Verified provider

Instance type

Instance type

t3.micro Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

All generations

Compare instance types

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

WebAppTier Create new key pair

Summary

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...read more
ami-01a00762f46d584a1

Virtual server type (instance type)
t3.micro

Firewall (security group)
SG Public Web

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTemplate:

EC2 > Launch templates > Create launch template

▼ Network settings Info

Subnet Info

Don't include in launch template

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone Info

Don't include in launch template

Not applicable for EC2 Auto Scaling

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group

Create security group

Common security groups Info

Select security groups

Compare security group rules

SG Public Web sg-06218c2b2e902ec2

VPC: vpc-07b3d399b08105f00

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Advanced network configuration

▼ Summary

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...read more

ami-01a00762f46d584a1

Virtual server type (instance type)

t3.micro

Firewall (security group)

SG Public Web

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Create launch template

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1

Choose launch template

Step 2

Choose instance launch options

Step 3 - optional

Integrate with other services

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Choose launch template Info

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name

Enter a name to identify the group.

ASG-Web-Tier

Must be unique to this account in the current Region and no more than 255 characters.

Launch template Info

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

Web-Server-Launch-Template

Create a launch template

Version

Default (1)

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1

Choose launch template

Step 2

Choose instance launch options

Step 3 - optional

Integrate with other services

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Integrate with other services - optional Info

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing Info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

No load balancer

Attach to an existing load balancer

Attach to a new load balancer

Attach to an existing load balancer

Select the load balancers to attach

Choose from your load balancer target groups

Choose from Classic Load Balancers

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

TG-Web-Server | HTTP

Step 7
Review

Network info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-07b3d399b08105f00 (Guvi 3 tier Project)
10.0.0.0/16

[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

aps1-az1 (ap-south-1a) | subnet-098fe46a2f49231c0 (Public Web 1)
10.0.1.0/24

aps1-az3 (ap-south-1b) | subnet-0f4abe6b299eb53af (Public Web 2)
10.0.2.0/24

[Create a subnet](#)

Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy zone.

☐ **Reservations Then Balanced**
Prioritizes launching into Capacity Reservations, distributing across AZs with available reservations.

Step 1
Choose launch template

Step 2
Choose instance launch options

Step 3 - optional
Integrate with other services

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Configure group size and scaling - *optional* info

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size info
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

2

Scaling info
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity
2

Max desired capacity
4

Equal or less than desired capacity Equal or greater than desired capacity

Auto Scaling groups (1/1) info Last updated 1 minute ago

[Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

Search your Auto Scaling groups

☒	Name	Status - new	Launch template/configuration	Instances	Instance health - new	Desire
☒	ASG-Web-Tier	At desired capacity	Web-Server-Launch-Template Version D	2	2/2 Healthy	2

Auto Scaling group: ASG-Web-Tier

Details | Integrations | Automatic scaling | Instance management | **Instance refresh** | Activity | Monitoring | Tags

Active instance refresh info [Start rollback](#) [Cancel instance refresh](#) [Start instance refresh](#)

No active instance refresh

Start an instance refresh to perform rolling updates on the Auto Scaling group's instances. Only one instance refresh can be active at a time.

The screenshot shows a web browser window with the AWS Management Console. The address bar contains the URL: `ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateAutoScalingGroup:`. The page title is "Create Auto Scaling group | EC2". The browser's top bar includes links for "CloudShell", "Agent Toolkit for AWS", "Feedback", "Console Mobile App", and copyright information for Amazon Web Services. The bottom bar shows the AWS logo, a search bar, and the user's name "Kumaresan S".

[CloudShell](#)
[Agent Toolkit for AWS](#)
[Feedback](#)
[Console Mobile App](#)

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[Cookie preferences](#)

Step 3 - optional
Integrate with other services

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Load balancing [info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ **Attach to an existing load balancer**
Choose from your existing load balancers.

☐ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach

☒ **Choose from your load balancer target groups**
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

TG-App-Server | HTTP
Application Load Balancer: ALB-App-Server

VPC Lattice integration options [info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and

Choose launch template

Step 2
Choose instance launch options

Step 3 - optional
Integrate with other services

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Configure group size and scaling - optional [info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

2

Scaling [info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity
2
Equal or less than desired capacity

Max desired capacity
4
Equal or greater than desired capacity

Automatic scaling - optional

ASG-App-Tier created successfully

Auto Scaling groups (2) [Info](#)

Last updated less than a minute ago

[Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

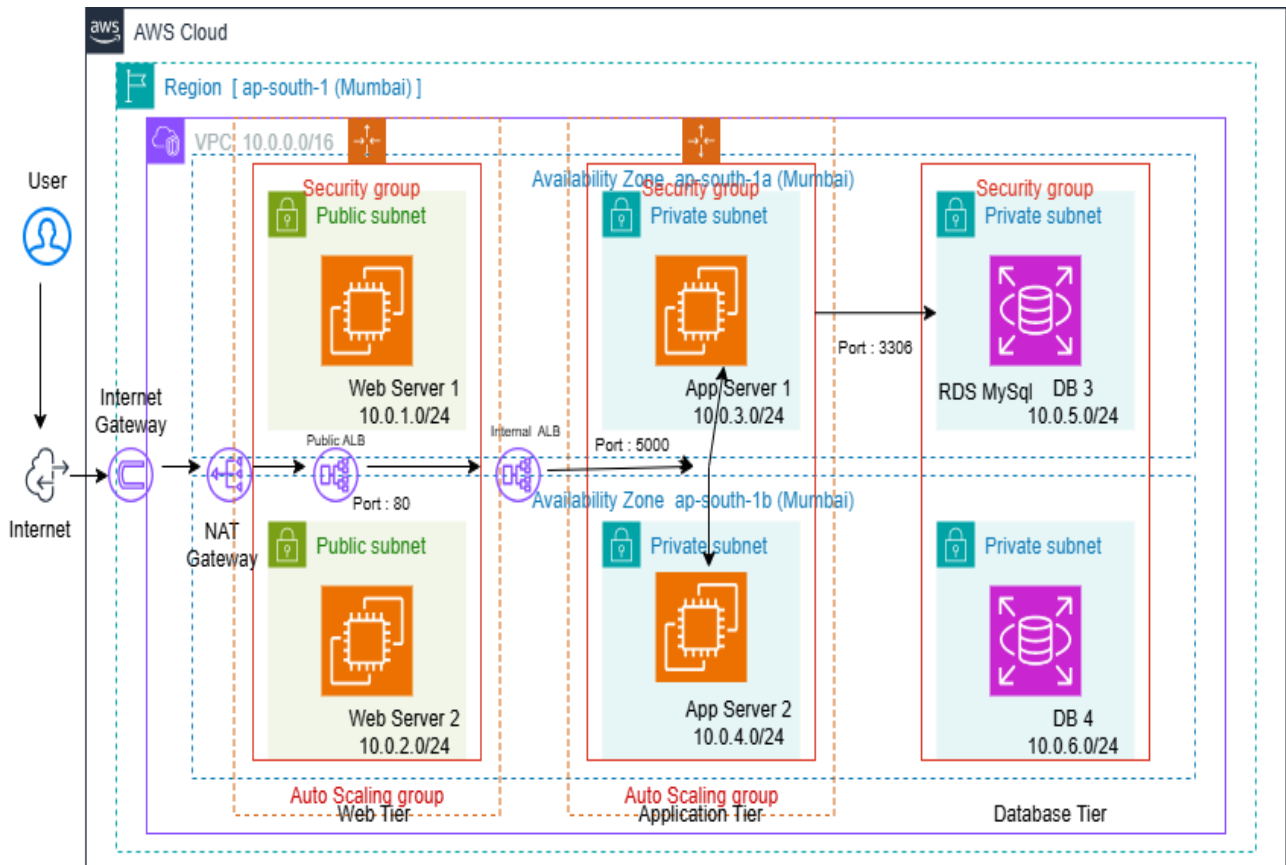
Search your Auto Scaling groups

☐	Name	Status - new	Launch template/configuration	Instances	Instance health - new	Des
☐	ASG-App-Tier	At desired capacity	App-Server-Launch-Template Version D	2	2/2 Healthy	2
☐	ASG-Web-Tier	At desired capacity	Web-Server-Launch-Template Version D	2	2/2 Healthy	2

0 Auto Scaling groups selected

Select an Auto Scaling group

AWS Architecture Diagram



This diagram shows a **3-tier web application deployed on AWS Mumbai Region (ap-south-1)**. The application uses a VPC with two Availability Zones for better availability.

- **Web Tier:** Two public subnets contain Web Server 1 and Web Server 2 running Nginx.
- **Application Tier:** Two private subnets contain App Server 1 and App Server 2 running the Flask application.
- **Database Tier:** Private database subnets are used for **Amazon RDS MySQL**.
- **Auto Scaling:** Web and application servers can be managed using Auto Scaling for scalability and fault tolerance.
- **Security:** The tiers are separated using public/private subnets and security groups.

SQL query output showing stored feedback records

SQL query output before auto scaling group:

```
$ mysql -h three-tier-mysql.cpukm8ci0ey5.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 284
Server version: 8.4.9 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE feedbackdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM feedback;
```

id	name	email	feedback	created
1	Test	t@t.com	Hello RDS!	2026-09-21 09:58:45
2	Test	t@t.com	Hello RDS!	2026-09-21 10:05:47
3	Abcd	abcd123.com	The session was good	2026-09-21 10:06:07
4	Kumaresan	test@example.com	Testing through Nginx	2026-09-21 12:49:57
5	Xyz	xyz@123.com	This session is very good	2026-09-21 12:50:21
6	Guvi	guviDevops@123.com	All should done the project	2026-09-21 13:03:01
7	Arun	arun@example.com	The feedback form is working correctly.	2026-09-21 16:11:01

```
7 rows in set (0.00 sec)

mysql>
```

alb-web-server-134995308.ap-south-1.elb.amazonaws.com

Feedback Form

RDS database connectivity test completed successfully

Submit

Feedback submitted successfully!

```
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM feedback;
```

id	name	email	feedback	created
1	Test	t@t.com	Hello RDS!	2026-09-21 09:58:45
2	Test	t@t.com	Hello RDS!	2026-09-21 10:05:47
3	Abcd	abcd123.com	The session was good	2026-09-21 10:06:07
4	Kumaresan	test@example.com	Testing through Nginx	2026-09-21 12:49:57
5	Xyz	xyz@123.com	This session is very good	2026-09-21 12:50:21
6	Guvi	guviDevops@123.com	All should done the project	2026-09-21 13:03:01
7	Arun	arun@example.com	The feedback form is working correctly.	2026-09-21 16:11:01

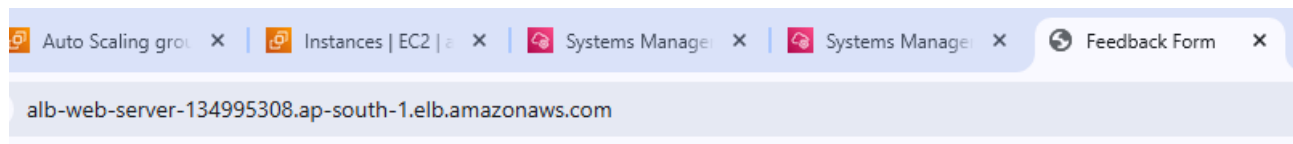
```
7 rows in set (0.00 sec)

mysql> SELECT * FROM feedback;
```

id	name	email	feedback	created
1	Test	t@t.com	Hello RDS!	2026-09-21 09:58:45
2	Test	t@t.com	Hello RDS!	2026-09-21 10:05:47
3	Abcd	abcd123.com	The session was good	2026-09-21 10:06:07
4	Kumaresan	test@example.com	Testing through Nginx	2026-09-21 12:49:57
5	Xyz	xyz@123.com	This session is very good	2026-09-21 12:50:21
6	Guvi	guviDevops@123.com	All should done the project	2026-09-21 13:03:01
7	Arun	arun@example.com	The feedback form is working correctly.	2026-09-21 16:11:01
8	Priya	priya@example.com	RDS database connectivity test completed successfully	2026-09-21 16:13:35

```
8 rows in set (0.00 sec)
```

SQL query output after auto scaling group:



Feedback Form

Feedback submitted successfully!

```
mysql> USE feedbackdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM feedback;
```

id	name	email	feedback	created
1	Test	t@t.com	Hello RDS!	2026-09-21 09:58:45
2	Test	t@t.com	Hello RDS!	2026-09-21 10:05:47
3	Abcd	abcd123.com	The session was good	2026-09-21 10:06:07
4	Kumaresan	test@example.com	Testing through Nginx	2026-09-21 12:49:57
5	Xyz	xyz@123.com	This session is very good	2026-09-21 12:50:21
6	Guvi	guviDevops@123.com	All should done the project	2026-09-21 13:03:01
7	Arun	arun@example.com	The feedback form is working correctly.	2026-09-21 16:11:01
8	Priya	priya@example.com	RDS database connectivity test completed successfully	2026-09-21 16:13:35
9	Cat	cat@gmail.com	enjoy your meal	2026-09-21 16:44:41

Check Feedback in MySQL

The feedback submitted through the web application is stored in the **MySQL RDS database**. The following commands are used to connect to the database and verify the stored feedback records.

```
mysql -h <RDS-ENDPOINT> -u <USERNAME> -p
```

After connecting to MySQL:

```
USE feedbackdb;
```

To check the feedback table:

```
SHOW TABLES;
```

To display all submitted feedback:

```
SELECT * FROM feedback;
```